

BERTastic at SemEval-2025 Task 10:

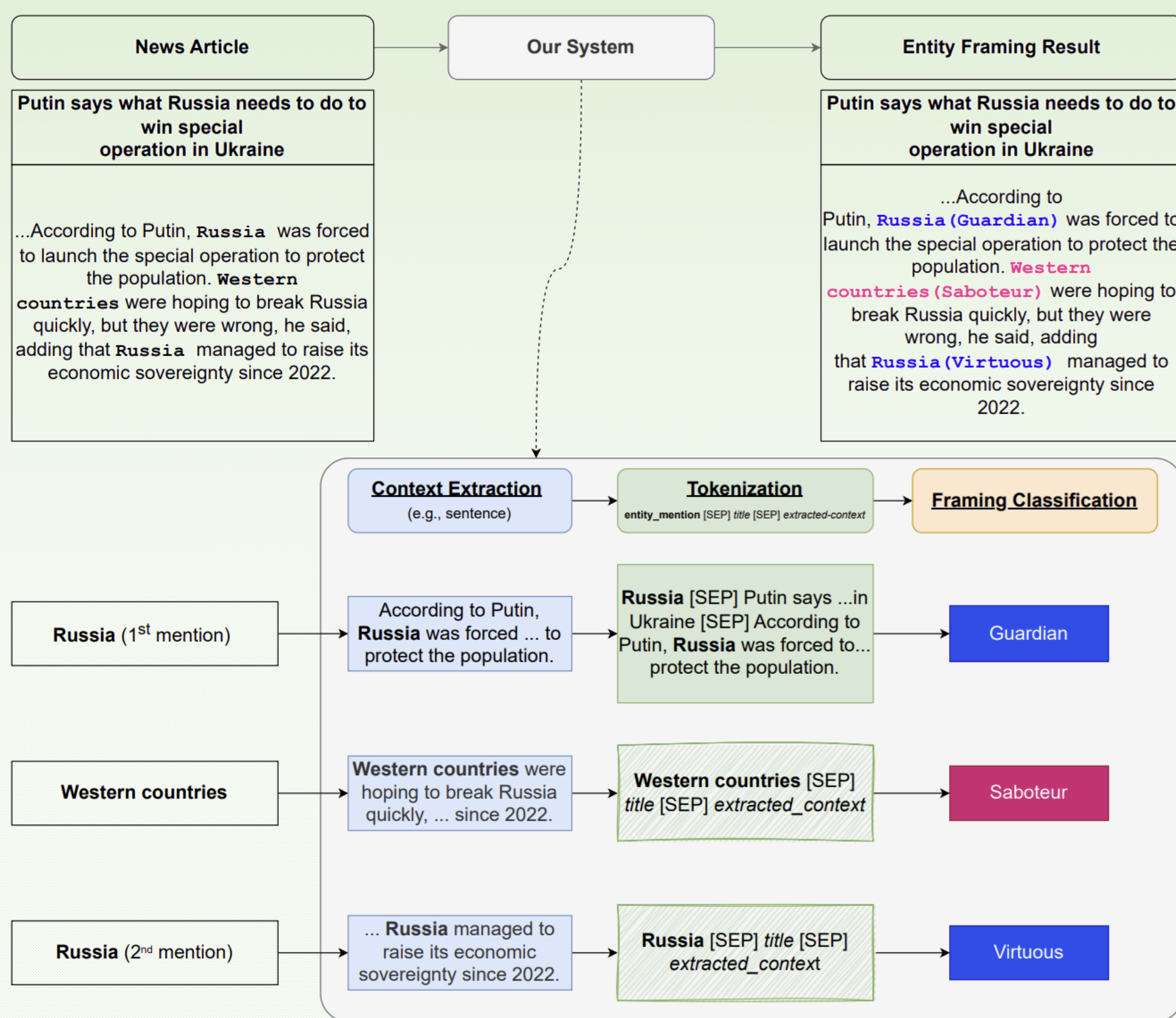
State-of-the-Art Accuracy in Coarse-Grained Entity Framing for Hindi News

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Contributions

- Demonstrated the strengths and trade-offs of zero-shot prompting vs. supervised fine-tuning for multilingual narrative analysis, informing future media bias system design:
 - Prompting: Explored **single-step**, and **least-to-most** hierarchical prompting strategies using GPT-4o.
 - Fine-tuning: Fine-tuned XLM-R across multiple granularities (sentence, paragraph, full text) and setups (monolingual vs. multilingual, main vs. fine-grained roles).
- Best System Performance:
 - Achieved **78.48%** main role accuracy on Hindi news—the highest on the official SemEval-2025 Task 10 leaderboard.
 - Reached 51.57% micro F1 for fine-grained role prediction across languages using sentence-level context.

Our System



Dual Approach Evaluation

Method	Lang.	Main Role		Fine Grained Role				Cost (USD)
		Accuracy	Balanced Accuracy	P	R	Micro F1	Macro F1	
Single-Step LLM Prompting	EN	.8346	.6756	.2692	.4632	.3405	.2171	0.7989
	BG	.8065	.7380	.3725	.5588	.4471	.3481	0.2751
	HI	.6327	.6247	.2753	.4000	.3262	.2196	2.4696
	PT	.7812	.7455	.5167	.6643	.5813	.2891	1.0200
	RU	.7558	.6719	.3939	.5843	.4706	.4644	0.7587
	All	.7030	.6957	.3211	.4726	.3824	.3103	5.3223
Multi-Step LLM Prompting	EN	.8031	.6799	.2887	.4118	.3394	.2383	0.5130
	BG	.8031	.6799	.4318	.5588	.4872	.3601	0.5130
	HI	.6367	.6284	.2676	.2868	.2769	.1771	1.4581
	PT	.8125	.7882	.3895	.2643	.3149	.2498	0.5634
	RU	.7442	.6680	.4118	.4719	.4398	.3774	0.4769
	All	.7053	.7017	.3051	.3294	.3168	.2765	3.1852
XLM-R	EN	0.6889	0.5276	.1854	.2828	.2240	.1327	—
	BG	0.7333	0.5791	.3030	.3030	.3030	.1349	—
	HI	0.7025	0.7046	.3234	.4951	.3912	.2043	—
	PT	0.8957	0.8840	.6259	.7480	.6815	.2040	—
	RU	0.8000	0.7604	.4831	.4886	.4859	.2364	—
	All	0.7529	0.7553	.3649	.4985	.4213	.2392	—

Experiments

Accuracy – Main Role – Hindi

Rank	Team	Main Role Acc.
1	BERTastic (Ours)	78.48%
2	QUST	76.90%
3	DEMON	75.63%
4	gowithnlp	71.52%
5	Dhananjaya	70.89%
6	PATeam	69.62%
7	FromProblemImportSolve	66.77%
8	Fane	65.51%
9	LATeHIMAS	63.61%
10	DUTIR	59.81%
11	Cimba	47.15%
12	LTG	37.66%
13	HowardUniversityAI4PC	35.44%
14	TartanTritons	32.91%
15	Baseline	32.28%

Monolingual vs. Multilingual

(a) Monolingual setting					(b) Multilingual setting				
Lang.	P	R	Micro F1	Macro F1	Lang.	P	R	Micro F1	Macro F1
EN	.1032	.1313	.1156	.0435	All	.3649	.4985	.4213	.2392
BG	.1056	.5758	.1784	.0505	EN	.1854	.2828	.2240	.1327
HI	.3424	.4495	.3887	.1740	BG	.3030	.3030	.3030	.1349
PT	.6124	.6423	.6270	.1505	HI	.3234	.4951	.3912	.2043
RU	.1077	.5227	.1786	.0437	PT	.6259	.7480	.6815	.2040
					RU	.4831	.4886	.4859	.2364

Context Granularity Evaluation

Train	Context	Main Role		Fine Grained Role			
		Accuracy	Balanced Accuracy	P	R	Micro F1	Macro F1
M	DOC	.6010	.5904	—	—	—	—
	PAR	.7379	.7385	—	—	—	—
	SEN	.7179	.7123	—	—	—	—
F	DOC	.7229	.7235	.3495	.4446	.3913	.2306
	PAR	.7529	.7553	.3649	.4985	.4213	.2392
	SEN	.7496	.7503	.4195	.4492	.4339	.2529



Entity Framing



Shared Task



Leaderboard